

TP n°4 Créer son Image « Portfolio » avec Docker

3 Création d'une image docker

```
#Creation et lancement du conteneur monWeb
#
docker run -d --name monWeb --restart=always -p 90:80 --network rsxsae23 webdevops/php-nginx
#Creation fichier index
#
docker exec -ti monWeb sh -c 'echo \"Bienvenue chez DIALLO-Thierno\" > /app/index.php'
```

3.1 Création d'une image à partir d'un conteneur

```
#Transformation du conteneur cree en image et list des images
#
docker commit monWeb im_web
docker image ls
#Suppression du conteneur monWeb et recreation d'un conteneur mon_web a partir de l'image cree precedement
#
docker rm monWeb
docker create --name mon_web -p 90:80 im_web
docker start mon_web
#
```

3.2 Création d'une image à partir d'un dockerfile

```
#Creation d'une image a partir d'un Dockerfile
mkdir rep_web_d
cd rep_web_d
touch photo.jpg
touch Dockerfile
echo "FROM webdevops/php-nginx
COPY photo.jpg /app/photo.jpg
RUN echo \"<p>Coucou c'est NOM</p>\" > /app/index.php
RUN echo \"<img src='photo.jpg'>\" > /app/index.php
EXPOSE 80" >> Dockerfile
cd ..
docker build -t img_web_d rep_web_d
docker image ls
docker create --name mon_web_dockfi -p 90:80 img_web_d
docker start mon_web_dockfi
```

4 Publication d'une image docker sur un dépôt privé

4.1 Installation du dépôt privé

```
user7@cli:~/sae203$ docker run -d -p 5000:5000 --restart always --name registry registry:3
Unable to find image 'registry:3' locally
3: Pulling from library/registry
42b9845e5d5c: Pull complete
d5a24b5445d2: Pull complete
d1dd084e87d: Pull complete
2c362d1d33f0: Pull complete
097321a3f414: Download complete
7652fa2b866e: Download complete
Digest: sha256:85347ed2ecde64161c7a4788a4d7d3dcc9d6f86f7be95834022e3c6a423a945a
Status: Downloaded newer image for registry:3
160b966a82e7a2e44ab36c4d9fef004ab3dd243f730c8cb74f499cb52413f11c
```

4.2 Stockage de l'image

```
user7@cli:~/sae203$ docker tag img_web_diallo localhost:5000/img_web_diallo
user7@cli:~/sae203$ docker image ls
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
alpine:latest	5b10f432ef3d	13.1MB	3.95MB	
hello-world:latest	f9078146db2e	25.9kB	9.49kB	
in_web:latest	75b3ac44111d	782MB	185MB	
img_web_diallo:latest	9b008e9df078	780MB	184MB	
localhost:5000/img_web_diallo:latest	9b008e9df078	780MB	184MB	
marladb:latest	e0236fc6386e	467MB	111MB	U
phpmyadmin:latest	0480a7583f87	821MB	197MB	U
portainer/helper-reset-password:latest	ef81b3973cca	83.4MB	25.5MB	
portainer/portainer-ce:latest	1ae8e65d50ca	240MB	59MB	U
registry:3	85347ed2ecde	84.5MB	20.5MB	U
webdevops/php-nginx:latest	d767e533e0fc	780MB	184MB	
wordpress:latest	054e61133439	1.08GB	275MB	U

```
user7@cli:~/sae203$ docker push localhost:5000/img_web_diallo
Using default tag: latest
The push refers to repository [localhost:5000/img_web_diallo]
65538e0c8933: Pushed
b03daebab708: Pushed
b6a6a4569330: Pushed
4c8b85719399: Pushed
5bed26d33875: Pushed
20d592fa47f1: Pushed
930bda195c84: Pushed
78bf9a5ad49e: Pushed
f70f19977fc5: Pushed
5131985698ec: Pushed
13b194aa3b52: Pushed
a13f330dd8cf: Pushed
d43f884b2c0e: Pushed
bf1db95ead3f: Pushed
d43c0a642c86: Pushed
30611fcc1b5e: Pushed
d3f32720a7f3: Pushed
f11b29a9c730: Pushed
latest: digest: sha256:9b008e9df078a28fb14bf9bebf0cfabb66a468723198aa8bffe6c787a3e9cd1d size: 856
```

4.3 Récupération d'une image sur un registre privé

```
user7@cli: /sae203
user7@cli: /sae203$ sudo vi /etc/docker/daemon.json
[sudo] Mot de passe de user7 :
user7@cli: /sae203$ sudo systemctl daemon-reload
user7@cli: /sae203$ sudo systemctl restart docker
user7@cli: /sae203$

6 mai 15:45

mh@monKanban: ~
" insecure-registries": ["192.168.203.7:5000"]

6 mai 15:49
Analyseur d'utilisation des disques A l'instant
Espace disque faible sur « Racine du système de fichiers »
Le volume « Racine du système de fichiers » n'a plus que 0 octet
d'espace disque disponible.
Examiner Ignorer

mh@monKanban: $ ping 192.168.203.7
PING 192.168.203.7 (192.168.203.7) 56(84) bytes of data:
64 bytes from 192.168.203.7: icmp_seq=1 ttl=64 time=1.02 ms
^C
--- 192.168.203.7 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 1.023/1.023/1.023/0.000 ms
mh@monKanban: $ curl http://192.168.203.7:5000/v2/_catalog
{"repositories":["img_web_diallo"]}
mh@monKanban: $ docker pull 192.168.203.7:5000/img_web_diallo
Using default tag: latest
latest: Pulling from img_web_diallo
a13f338dd8cf: Pull complete
5131985698ec: Pull complete
5bed26d33875: Pull complete
f11b29a9c730: Pull complete
930bda195c84: Pull complete
78bf9a5ad49e: Pull complete
4c8b85719399: Pull complete
30611fcc1b5e: Pull complete
f70f19977fc5: Pull complete
d3f32720a7f3: Pull complete
65330ebec9933: Pull complete
d43f884b2c0e: Pull complete
20d592fa47f1: Pull complete
b03daebab708: Pull complete
bf1db95ead3f: Pull complete
d6a6a4569338: Pull complete
d43c0a642e06: Pull complete
13b194aa3b52: Download complete
Digest: sha256:9b008e9df070a28fb14bf9bebf0cfabb66a468723198aa8bffe6c787a3e9cd1d
Status: Downloaded newer image for 192.168.203.7:5000/img_web_diallo:latest
192.168.203.7:5000/img_web_diallo:latest
mh@monKanban: $
```

5 Création d'une image Portfolio

5.1 Recensement des commandes

```
user7@cli: /sae203$ docker run -ti --name test -p 82:82 debian /bin/bash
Unable to find image 'debian:latest' locally
latest: Pulling from library/debian
3b32e3bb7338: Pull complete
46cee010f6c: Download complete
Digest: sha256:35b8ff74ead4880f22090b617372daff0ccae742eb5674455d542bef71ef1999
Status: Downloaded newer image for debian:latest
root@bb4891272cf7:/#

#apt update
#apt install apache2 php php-xml php-mbstring php-intl php-mysql zip git -y
#php -r "copy('https://getcomposer.org/installer', '/root/composer-setup.php');"
#php /root/composer-setup.php --install-dir="/usr/bin"
```

```
root@bb4891272cf7:/# apt install apache2 php php-xml php-mbstring php-intl php-mysql zip git
Installing:
  apache2  git  php  php-intl  php-mbstring  php-mysql  php-xml  zip

Installing dependencies:
  adduser          libargon2-1      libgdbm6t64      libldap-common   librtmp1          libxcb1          perl-modules-5.40  publicsuffix
  apache2-bin      libbrotli1       libgnutls30t64   libldap2         libsasl2-2        libxdmcp6        php-common         sensible-utils
  apache2-data     libcbor0.10      libgpg-error-l10n liblua5.4-0      libsasl2-modules  libxext6         php8.4            ssl-cert
  apache2-utils    libcom-err2      libgpg-error0     libncursesw6     libsasl2-modules-db libxml2          php8.4-cli        ucf
  ca-certificates  libcurl3t64-gnutls libgpm2          libnghttp2-14    libssodium23      libxmuu1         php8.4-common     unzip
  git-man          libcurl4t64      libgssapi-krb5-2  libnghttp3-9     libssh2-1t64      libxslt1.1       php8.4-intl       xauth
  krb5-locales     libedit2         libicu76         libnghttp2-16    libtasn1-6         linux-sysctl-defaults php8.4-mbstring
  less            liberror-perl    libidn2-0        libngtcp2-crypto-gnutls8 libtext-charwidth-perl media-types      php8.4-mysql
  libapache2-mod-php8.4 libexpat1        libjansson4      libngtcp2-crypto-gnutls8 libtext-wrap18n-perl netbase         php8.4-opcache
  libapr1t64       libffi8          libk5crypto3      libonig5         libtext-wrap18n-perl libunistring5    openssh-client   php8.4-readline
  libaprutil1-dbd-sqlite3 libffi8          libkeyutils1      libperl5.40      libx11-6           openssl          php8.4-xml
  libaprutil1-ldap libgcrypt20       libkrb5-3         libproc2-0       libx11-data        patch           procps
  libaprutil1t64   libgdbm-compat4t64 libkrb5support0   libpsl5t64       libxau6            perl            psmisc

Suggested packages:
  liblocale-gettext-perl  ufw          gitk          rng-tools      libsasl2-modules-gssapi-mit  libpam-ssh      libterm-readline-gnu-perl
  cron                   www-browser  gitweb        gdbm-l10n      | libsasl2-modules-gssapi-heimdall  monkeysphere   | libterm-readline-perl-perl
  quota                  gettext-base  git-cvs       gnutls-bin    libsasl2-modules-ldap        ssh-askpass    make
  apache2-doc            git-doc       git-mediatool  gpm           libsasl2-modules-otp         ed              libtap-harness-archive-perl
  apache2-suexec-pristine git-email     git-svn       krb5-doc      libsasl2-modules-sql        diffutils-doc  perl-doc
  | apache2-suexec-custom git-gui       php-pear      krb5-user     keychain

Summary:
  Upgrading: 0, Installing: 105, Removing: 0, Not Upgrading: 0
  Download size: 52.1 MB
  Space needed: 224 MB / 18.8 GB available

root@241899a38b38:/# php -r "copy('https://getcomposer.org/installer', '/root/composer-setup.php');"
php /root/composer-setup.php --install-dir="/usr/bin"
All settings correct for using Composer
Downloading...

Composer (version 2.9.8) successfully installed to: /usr/bin/composer.phar
Use it: php /usr/bin/composer.phar

root@241899a38b38:/#
```

- Récupération de votre site

Clonez votre dépôt git portfolio dans le répertoire /var/www/html/site

```
root@241899a38b38:/# mkdir /var/www/html/site
root@241899a38b38:/# cd /var/www/html/site
root@241899a38b38:/var/www/html/site# git clone https://github.com/mhd7pf/myPortfolio.git
Cloning into 'myPortfolio'...
remote: Enumerating objects: 375, done.
remote: Counting objects: 100% (22/22), done.
remote: Compressing objects: 100% (22/22), done.
Receiving objects: 12% (45/375), 34.03 MiB | 6.98 MiB/s
```

- Installation des dépendances

Demander à composer de télécharger les dépendances de votre site

```
root@241899a38b38:/var/www/html/site# cd myPortfolio/
root@241899a38b38:/var/www/html/site/myPortfolio# composer.phar -n update && composer.phar -n install
Composer is operating significantly slower than normal because you do not have the PHP curl extension enabled.
Loading composer repositories with package information
Updating dependencies
Lock file operations: 1 install, 94 updates, 0 removals
- Upgrading doctrine/collections (2.4.0 => 2.6.0)
- Upgrading doctrine/dbal (3.10.3 => 3.10.5)
- Upgrading doctrine/deprecations (1.1.5 => 1.1.6)
- Upgrading doctrine/doctrine-bundle (2.18.1 => 2.18.2)
- Upgrading doctrine/doctrine-migrations-bundle (3.6.0 => 3.7.0)
- Upgrading doctrine/event-manager (2.0.1 => 2.1.1)
- Upgrading doctrine/instantiator (2.0.0 => 2.1.0)
- Upgrading doctrine/migrations (3.9.4 => 3.9.7)
- Upgrading doctrine/orm (3.5.7 => 3.6.5)
- Upgrading doctrine/persistence (4.1.1 => 4.2.0)
- Upgrading doctrine/sql-formatter (1.5.3 => 1.5.4)
- Upgrading monolog/monolog (3.9.0 => 3.10.0)
- Upgrading nikitic/php-parser (v5.6.2 => v5.7.0)
- Upgrading phpdocumentor/reflection-docblock (5.6.3 => 5.6.7)
- Upgrading phpdocumentor/type-resolver (1.10.0 => 1.12.0)
- Upgrading phpstan/phpdoc-parser (2.3.0 => 2.3.2)
- Upgrading phpunit/php-code-coverage (12.4.0 => 12.5.6)
- Upgrading phpunit/php-file-iterator (6.0.0 => 6.0.1)
- Upgrading phpunit/phpunit (12.4.3 => 12.5.25)
- Upgrading sebastian/comparator (7.1.3 => 7.1.6)
- Upgrading sebastian/environment (8.0.3 => 8.1.0)
- Upgrading symfony/asset (v7.3.0 => v7.3.10)
- Upgrading symfony/asset-mapper (v7.3.5 => v7.3.10)
- Upgrading symfony/browser-kit (v7.3.6 => v7.3.10)
- Upgrading symfony/cache (v7.3.6 => v7.4.10)
- Upgrading symfony/cache-contracts (v3.6.0 => v3.7.0)
- Upgrading symfony/clock (v7.3.0 => v7.4.8)
- Upgrading symfony/config (v7.3.6 => v7.4.10)
- Upgrading symfony/console (v7.3.6 => v7.3.10)
- Upgrading symfony/dependency-injection (v7.3.6 => v7.4.10)
- Upgrading symfony/deprecation-contracts (v3.6.0 => v3.7.0)
```

- Configuration de apache

Nous n'avons pas installé l'utilitaire symfony qui vous permettait de lancer votre site web : c'est normal, nous allons utiliser apache comme serveur web et pas symfony.

On peut installer les paquets lynx (navigateur web en mode console)

```
root@241899a38b38:/var/www/html/site/myPortfolio# apt install lynx -y
Installing:
  lynx

Installing dependencies:
  bzip2 file libmagic-mgc libmagic1t64 lynx-common mailcap xz-utils

Suggested packages:
  bzip2-doc

Summary:
  Upgrading: 0, Installing: 8, Removing: 0, Not Upgrading: 9
  Download size: 3058 kB
  Space needed: 18.3 MB / 31.7 GB available

# Déclaration de l'un VirtualHost écoutant sur le port 80
<VirtualHost *:80>

    # Email de l'administrateur du site
    ServerAdmin thierno.nouhamadou.d.diallo@etu.univ-st-etienne.fr

    # Nom du serveur (localhost dans un conteneur)
    ServerName localhost

    # Répertoire racine du site Symfony (le dossier public)
    DocumentRoot /var/www/html/site/myPortfolio/public

    # Configuration des permissions sur le dossier public
    <Directory /var/www/html/site/myPortfolio/public>

        # Désactive les .htaccess (on utilise la conf Apache)
        AllowOverride None

        # Ancienne syntaxe Apache 2.2 (encore acceptée)
        Order Allow,Deny
        Allow from All

        # Active le moteur de réécriture si le module est chargé
        <IfModule mod_rewrite.c>
            RewriteEngine On

            # Si le URL contient une extension (ex: .css, .js, .png)
            RewriteCond %{REQUEST_URI} \.+$

            # mais que ce n'est pas un fichier .html
            RewriteCond %{REQUEST_URI} !\.html$

            # Alors on ne réécrit pas
            RewriteRule .* - [L]

            # Sinon, on redirige tout vers index.php (Symfony)
            RewriteRule ^(.*)$ index.php [QSA,L]
        </IfModule>

root@241899a38b38:/var/www/html/site/myPortfolio# vim /etc/apache2/sites-available/site.conf
root@241899a38b38:/var/www/html/site/myPortfolio# a2ensite site.conf
a2dissite 000-default.conf
a2enmod rewrite
Enabling site site.
To activate the new configuration, you need to run:
  service apache2 reload
Site 000-default disabled.
To activate the new configuration, you need to run:
  service apache2 reload
Enabling module rewrite.
To activate the new configuration, you need to run:
  service apache2 restart
root@241899a38b38:/var/www/html/site/myPortfolio# /usr/sbin/service apache2 start
Starting Apache httpd web server: apache2AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 172.17.0.3. Set the 'ServerName' directive globa
lly to suppress this message
root@241899a38b38:/var/www/html/site/myPortfolio#
```

5.2 Ajustement du site web

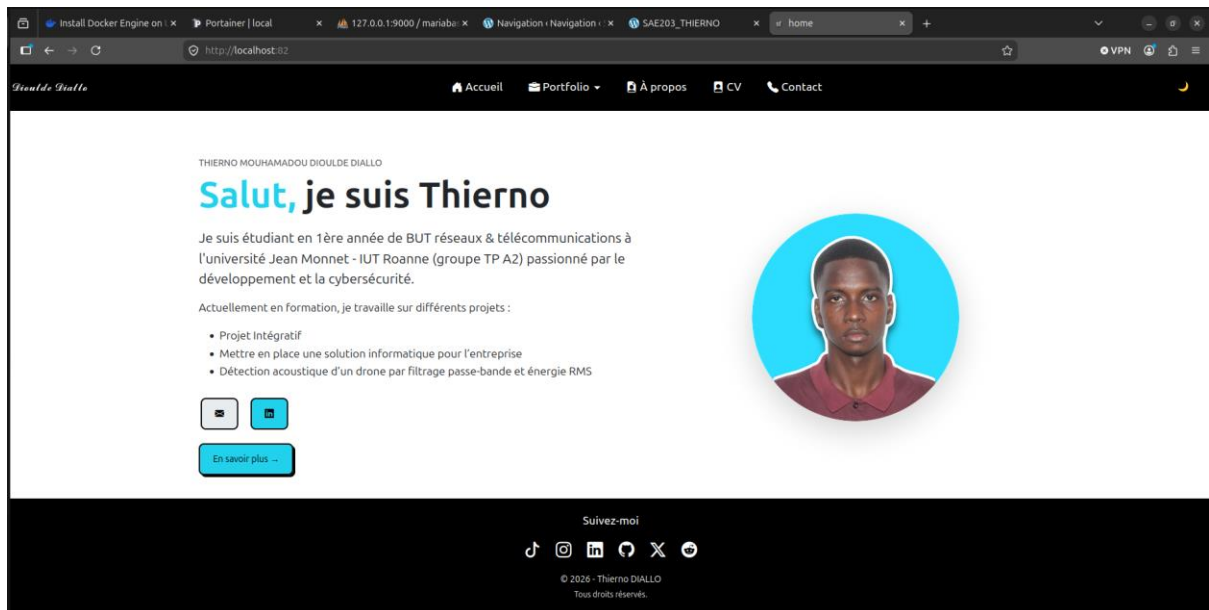
```
root@241899a38b38:/var/www/html/site/myPortfolio/public# cat index.php
<?php

use App\Kernel;

require_once dirname(__DIR__).'/vendor/autoload_runtime.php';

return function (array $context) {
    return new Kernel($context['APP_ENV'], (bool) $context['APP_DEBUG']);
};
root@241899a38b38:/var/www/html/site/myPortfolio/public#
```

J'ai rien eu à modifier dans le fichier index.php



5.3 Réalisation du Dockerfile

```
user17@tstdckr: ~/sae203/portefolio
FROM debian:latest

# Mise à jour + installation Apache, PHP et extensions
RUN apt update && apt install apache2 php php-xml php-mbstring php-intl php-mysql zip git -y

# Installation de Composer
RUN php -r "copy('https://getcomposer.org/installer', '/root/composer-setup.php');" \
    && php /root/composer-setup.php --install-dir="/usr/bin"

# Installation de Lynx (pour tester)
RUN apt update && apt install lynx -y

# Création du dossier du site
RUN mkdir -p /var/www/html/site

# Définition du répertoire de travail
WORKDIR /var/www/html/site

# Clone du dépôt Git
RUN git clone https://github.com/nhd7pf/myPortfolio.git .

# Installation des dépendances Symfony
RUN composer.phar -n update && composer.phar -n install

# Copie du fichier site.conf dans l'image
COPY site.conf /etc/apache2/sites-available/site.conf

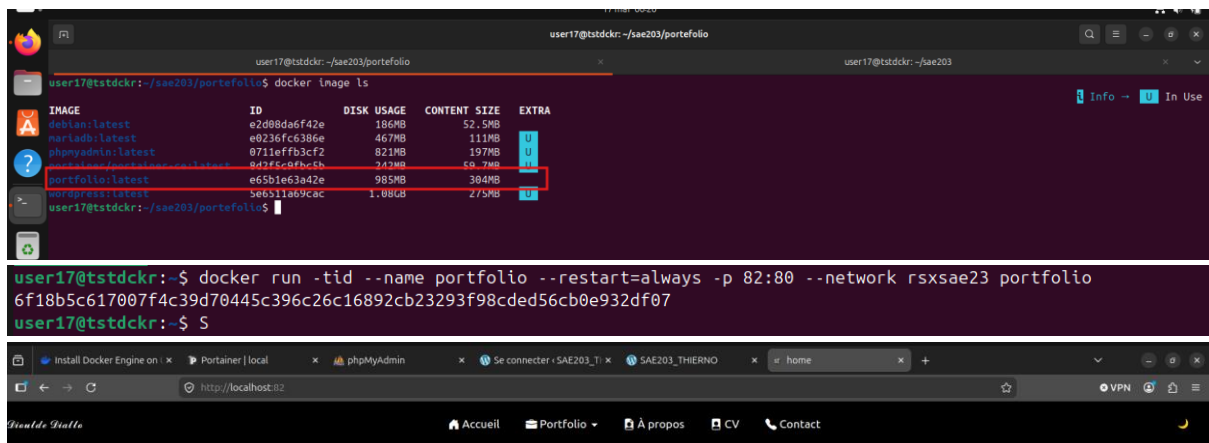
# Activation du VirtualHost + rewrite + désactivation du défaut
RUN a2enconf site.conf && a2dissite 000-default.conf && a2enmod rewrite

# Exposition du port 80 dans l'image
EXPOSE 80

# Lancement d'Apache + ouverture d'un bash pour garder le conteneur actif
CMD /usr/sbin/service apache2 start & bash

user17@tstdckr: ~/sae203/portefolio
user17@tstdckr: ~/sae203/portefolio$ vim Dockerfile
user17@tstdckr: ~/sae203/portefolio$ docker cp test:/etc/apache2/sites-available/site.conf .
Successfully copied 1.29kB (transferred 3.07kB) to /home/user17/sae203/portefolio/.
user17@tstdckr: ~/sae203/portefolio$ ls
Dockerfile  site.conf
user17@tstdckr: ~/sae203/portefolio$

user17@tstdckr: ~/sae203/portefolio$ docker build -t portfolio .
[+] Building 52.0s (5/14)
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 1.17kB
=> [internal] load metadata for docker.io/library/debian:latest
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [ 1/10] FROM docker.io/library/debian:latest@sha256:e2d08da6f42ef4b09b165d55528a12727eed8248dc9edf888e3ec07e18ef9da
=> => resolve docker.io/library/debian:latest@sha256:e2d08da6f42ef4b09b165d55528a12727eed8248dc9edf888e3ec07e18ef9da
=> [internal] load build context
=> => transferring context: 1.13kB
=> [ 2/10] RUN apt update && apt install apache2 php php-xml php-mbstring php-intl php-mysql zip git -y
=> => # debconf: unable to initialize frontend: Readline
=> => # debconf: (This frontend requires a controlling tty.)
=> => # debconf: falling back to frontend: Teletype
=> => # debconf: unable to initialize frontend: Teletype
=> => # debconf: (This frontend requires a controlling tty.)
=> => # debconf: falling back to frontend: Noninteractive
docker:default 51.5s
```



THIERNO MOUHAMADOU DIOULE DIALLO

Salut, je suis Thierno

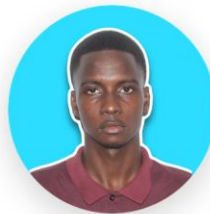
Je suis étudiant en 1ère année de BUT réseaux & télécommunications à l'université Jean Monnet - IUT Roanne (groupe TP A2) passionné par le développement et la cybersécurité.

Actuellement en formation, je travaille sur différents projets :

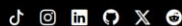
- Projet Intégré
- Mettre en place une solution informatique pour l'entreprise
- Détection acoustique d'un drone par filtrage passe-bande et énergie RMS



[En savoir plus ...](#)



Suivez-moi



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CMD :

Pour cree image : `docker build -t portfolio .`

Pour lancer conteneur :

`docker run -tid --name portfolio --restart=always -p 82:80 --network rsxsae23 portfolio`